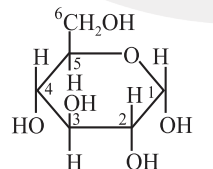
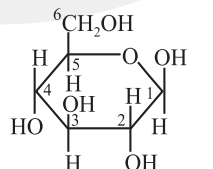


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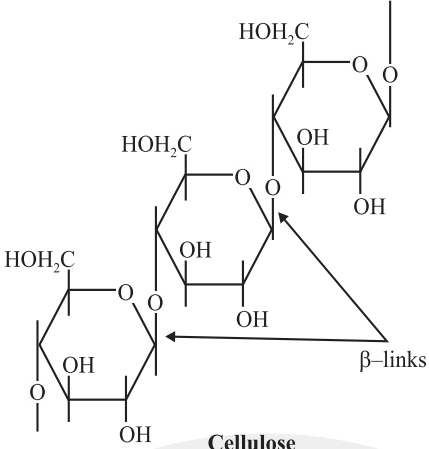
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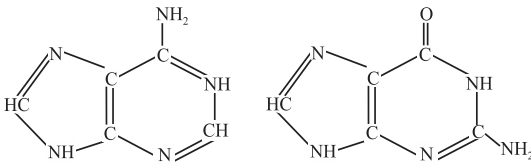
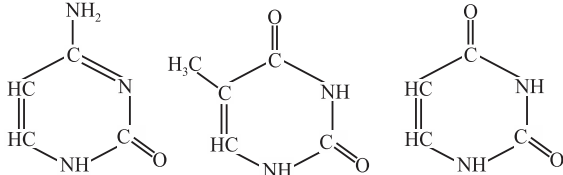
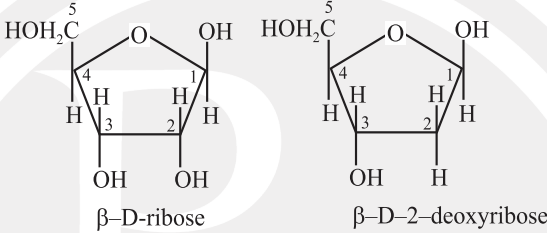
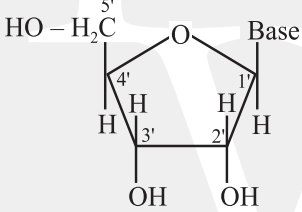
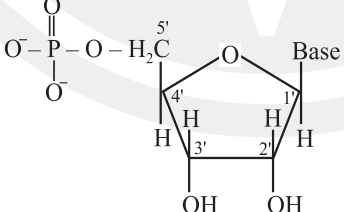
Biomolecules

Sr. No.	Concept	Description	Other information
1.	Carbohydrates (a) Monosaccharides (b) Oligosaccharides (c) Polysaccharides	$(CH_2O)_n$ Cannot be hydrolysed Yield 2-10 monosaccharides Yield more than 10 monosaccharides on hydrolysis	Examples Glucose Maltose Starch
2.	Sugar (a) Reducing sugars (b) Non-reducing sugars	Glucose, Fructose, Galactose Sucrose	
3.	Glucose (a) Structure (i) Fischer structure (ii) Cyclic structure (iii) Reactions	It is a monosaccharide with molecular formula $C_6H_{12}O_6$. (i) Fischer structure $ \begin{array}{c} \text{CHO} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{HO}-\text{C}-\text{H} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{CH}_2\text{OH} \\ \text{D-(+)-glucose} \end{array} $ (ii) Cyclic structure  $\alpha\text{-D-(+)-Glucopyranose}$  $\beta\text{-D-(+)-Glucopyranose}$ (iii) Reactions $ \begin{array}{c} \text{CHO} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array} \xrightarrow{\text{HI}, \Delta} \text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3 $ (<i>n</i> - Hexane)	Six carbon atoms linked in straight chain

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		$ \begin{array}{c} \text{CHO} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array} \xrightarrow{\text{NH}_2\text{OH}} \begin{array}{c} \text{CH=N-OH} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array} \quad \begin{array}{c} \text{CHO} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array} \xrightarrow{\text{HCN}} \begin{array}{c} \text{CH} \\ / \quad \backslash \\ \text{CN} \quad \text{OH} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array} $ $ \begin{array}{c} \text{CHO} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array} \xrightarrow{\text{Acetic anhydride}} \begin{array}{c} \text{CHO} \\ \\ (\text{CH-O-C-CH}_3)_4 \\ \quad \quad \quad \text{O} \\ \text{CH}_2\text{-O-C-CH}_3 \quad \quad \quad \text{O} \end{array} $ $ \begin{array}{c} \text{CHO} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array} \xrightarrow{\text{Br}_2 \text{ water}} \begin{array}{c} \text{COOH} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array} \quad \text{Gluconic acid} $ $ \begin{array}{c} \text{CHO} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array} \xrightarrow{\text{Oxidation}} \begin{array}{c} \text{COOH} \\ \\ (\text{CHOH})_4 \\ \\ \text{COOH} \end{array} \quad \begin{array}{c} \text{COOH} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array} \xleftarrow{\text{Oxidation}} \begin{array}{c} \text{COOH} \\ \\ (\text{CHOH})_4 \\ \\ \text{CH}_2\text{OH} \end{array} $ Saccharic acid Gluconic acid	Confirm presence of carbonyl group Confirm presence of five -OH group Indicate aldehyde group Indicate presence of primary alcoholic group
4.	Fructose (a) Structure (i) Fischer structure (ii) Cyclic structure	Fructose also has the molecular formula $\text{C}_6\text{H}_{12}\text{O}_6$. $ \begin{array}{c} \text{CH}_2\text{OH} \\ \\ \text{C=O} \\ \\ \text{HO}-\text{C}-\text{H} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{CH}_2\text{OH} \end{array} $ $\begin{array}{c} \text{HOH}_2\text{C}^6 \quad \text{O} \quad \text{CH}_2\text{OH}^1 \\ \quad \quad \quad \\ \text{H}^5 \quad \text{OH}^2 \\ \quad \quad \quad \\ \text{H}^4 \quad \text{OH}^3 \\ \quad \quad \quad \\ \text{OH} \quad \text{H} \end{array}$ $\alpha\text{-D-(-)-Fructofuranose}$ $\begin{array}{c} \text{HOH}_2\text{C}^6 \quad \text{O} \quad \text{OH} \\ \quad \quad \quad \\ \text{H}^5 \quad \text{OH}^2 \\ \quad \quad \quad \\ \text{H}^4 \quad \text{CH}_2\text{OH}^1 \\ \quad \quad \quad \\ \text{OH} \quad \text{H} \end{array}$ $\beta\text{-D-(-)-Fructofuranose}$	

Sr. No.	Concept	Description	Other information
5.	Sucrose	$\text{C}_{12}\text{H}_{22}\text{O}_{11} + \text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_{12}\text{O}_6 + \text{C}_6\text{H}_{12}\text{O}_6$ Sucrose D-(+)-Glucose D-(-)-Fructose Glycosidic linkage Sucrose	Invert sugar
6.	Maltose	(I) Maltose (II) α-D-Glucose α-D-Glucose	
7.	Lactose	β-D-Galactose β-D-Glucose Lactose	
8.	Starch	Starch is a polymer of α-glucose and consists of two components— Amylose and Amylopectin α-Link α-Link Amylose α-Link Branch at C_6 α-Link Amylopectin	

Sr. No.	Concept	Description	Other information
9.	Cellulose	 Cellulose	Cellulose composed of β -D-glucose
10.	Amino acids (a) Zwitter ion structure	Amino acids contain amino ($-\text{NH}_2$) and carboxyl ($-\text{COOH}$) functional groups. $\begin{array}{c} \text{R}-\text{CH}-\text{COOH} \\ \\ \text{NH}_2 \\ \alpha\text{-amino acid} \end{array}$ $\begin{array}{c} \text{R}-\text{CH}-\text{C}(=\text{O})-\text{O}-\text{H} \\ \\ \text{:NH}_2 \end{array} \rightleftharpoons \begin{array}{c} \text{R}-\text{CH}-\text{C}(=\text{O})-\text{O}^- \\ \\ \text{NH}_3^+ \\ \text{(Zwitter ion)} \end{array}$	R = side chain
11.	Protein (a) Fibrous proteins (b) Globular proteins	Keratin, myosin Insulin and albumin	These are macro molecules made up of amino acids joined by amide linkage [$-\text{CONH}-$] is here called as peptide linkage.
12.	Vitamins (a) Fat soluble vitamins: (b) Water soluble vitamins:	Vitamins A, D, E and K. Vitamins B and C.	
13.	Nucleic acids	These are biomolecules which are long chain polymers of nucleotides. Nuceloside = Base + Sugar Nucleotide = Base + Sugar + Phosphoric acid	

Sr. No.	Concept	Description	Other information
	(a) Base	 Adenine (A) Guanine (G)  Cytosine (C) Thymine (T) Uracil (U)	DNA → A, G, C, T RNA → A, G, C, U
	(b) Sugar	 β-D-ribose β-D-2-deoxyribose	
	(c) Structure		
	(i) Nucleoside	Base + sugar 	
	(ii) Nucleotide	Base + sugar + Phosphoric acid 	
	(d) Deoxyribonucleic acid (DNA)	Base + β - D - 2 - deoxyribose + Phosphoric acid	
	(e) Ribonucleic acid (RNA)	Base + β - D - ribose + Phosphoric acid	